Pelvic Viscera and Neurovasculature; Part 2

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Block: Endo/Uro/Repro

Thread: Anatomy

Conflict of Interest: Dr. Spudich has no conflicts of interest to disclose

Textbook

Moore's Clinically Oriented Anatomy, 9th ed; Go To: Chapter 6: Pelvis and Perineum

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the position, form, peritoneal coverings of the internal pelvic Describe the functional anatomy of the reproductive viscera concerned with Sperm maturation and transmission, including their anatomical relationships to abdominopelvic and perineal structures
2. Describe the functional anatomy of the reproductive viscera concerned with Oocyte maturation and transmission, including their anatomical relationships to abdominopelvic and perineal structures
3. Describe the sensory innervation of the uterus and vagina and how it impacts administration of anesthesia during childbirth

Study Questions

1. What is the most common site of ectopic gestation? And discuss.
    A. Abdominal cavity
    B. Ampulla of uterine tube
    C. Cervix of uterus
    D. Infundibulum of uterine tube
    E. Isthmus of uterine tube

2. Review the course of the vas deferens and discuss the normal site for the performance of a vasectomy.

3. A 32-year-old patient presents with gonorrhea and a tubo-ovarian abscess. Discuss the implications of passage of the bacteria through the vagina and uterus.

4. Compare and contrast the anesthesia options available during childbirth and describe the anatomic basis for each type.

I. Organs of Sperm Maturation and Transmission

1. The perineal components are
    a. Penis
    b. Testes
    c. Epididymis
    d. Ductus deferens
    e. Bulbourethral gland
2. Pelvic components
    a. Ductus
    b. Seminal gland
    c. Ejaculatory duct
    d. Prostate

J. Ductus deferens/Seminal Gland

1. Transmits sperm to prostatic urethra
2. Begins as a continuum of the epididymis
3. Runs in the spermatic cord and through the inguinal canal into the pelvis
4. Runs along the posterior wall of the bladder
5. The ampulla of the Vas meets up with the seminal gland
6. Seminal glands:
    a. Thick alkaline secretion mixing with sperms
7. Ejaculatory duct:
    a. Union of the ductus deferens and duct of seminal gland
    b. Traversing the prostate
    c. Opening on the seminal colliculus in the prostatic urethra

8. Prostatic urethra
    a. Seminal colliculus on the posterior wall
    b. Ejaculatory ducts empty here
    c. Utricle is an embryonic remnant

9. Prostate:
    a. Encapsulated fibromuscular gland
    b. Secretions empty into the seminal sinuses adjacent to colliculus
    c. Several Zones in the prostate
        i. Transitional zone – BPH
        ii. Peripheral zone – most of the glands felt in Rectal exam

10. Bulbo-urethral glands
    a. Embedded in the external urethral sphincter in the deep perineal space
2. The duct opening into the proximal spongy urethra

K. Vascular Supply of Ductus, Seminal Vesicle, and Prostate

11. The pelvic portions are all supplied by the superior and inferior vesicle arteries
12. Ductus in cord is supplied by the artery to the ductus a branch of superior vesicle artery
13. Venous blood collects in plexuses
    a. Prostatic
    b. Vesicle
14. Ductus in cord drains to testicular veins

L. Motor Innervation of Sperm Conducting organs

8. Sympathetic innervation from T12-L2/3 via lumbar splanchnics
9. Sympathetic stimulation cause:
10. Contraction of internal urethral sphincter to prevent retrograde ejaculation
11. Rapid peristaltic-like contraction of ductus deferens
12. Contraction and secretion of seminal glands and prostate
13. Parasympathetic motor stimulation from S2-4
14. Cavernous nerves to erectile bodies and causing penile erection
15. Visceral sensory goes to S2-4 as most is below pelvic pain line

M. Lymphatic Drainage of Organs in Spermatic Pathway

1. Testes drain to lumbar/aortic nodes
2. Internal iliac nodes drain
    a. Prostate
    b. Seminal gland
    c. Ductus
    d. Epididymis

N. Organs concerned with Oocyte maturation and transmission

1. Peritoneal reflections on viscera
2. Superior and superolateral surfaces of pelvic viscera are covered by peritoneum.
3. Most viscera are subperitoneal.
4. Reflection of peritoneum creates folds and fossae:
    i. Recto-vesical pouch
    ii. Recto-uterine pouch
    iii. Pararectal fossae
    iv. Recto-uterine folds: underlying fascial ligaments
5. Broad ligament of the uterus
    a. A double peritoneal fold covering the uterus
    b. Extending to lateral pelvic wall on each side
    c. Enclosing the uterus, uterine tubes, ligament of ovaries, round ligament of uterus, , NOT ovaries
6. Broad ligament has several regions
    a. Mesometrium
    b. Mesovarium
    c. Mesosalpinx
7. Uterine ligaments (passive support):
    a. Ligament of the ovary
    b. Suspensory ligament of the ovary
    c. Round ligament of the uterus

O. Ovaries

1. Almond-shaped gonads
    a. Oocytes development
    b. Reproductive hormones
2. Suspended in pelvis by mesovarium, a portion of the broad ligament of the uterus
    a. Covered by surface epithelium that becomes scarred due to discharge of oocytes at ovulation
    b. Suspensory ligament of ovary supplies blood
    c. Ligament of ovary attaches them to the uterus

P. Uterine tubes (Oviduct, fallopian tube)

1. Conducting oocyte discharged from the ovary
2. Site of fertilization
3. Extending from uterine horns and opening into the peritoneal cavity near the ovaries
4. Enclosed within mesosalpinx, a portion of the broad ligament
5. Four parts:
    a. Infundibulum: the funnel shaped distal end with abdominal ostium and fimbriae
    b. Ampulla: the widest and longest part, site of fertilization, usual site of ectopic tubal pregnancy
    c. Isthmus: thick-walled part entering the uterine horn
    d. Uterine part: intramural segment passing through the wall of the uterus

Q. Uterus

8. Relations of uterus
    a. Anteriorly: superior surface of the urinary bladder and vesico-uterine pouch
    b. Posteriorly: recto-uterine pouch and anterior surface of the rectum
    c. Laterally: broad ligament, cardinal ligaments, ureters lying inferior to the uterine arteries
9. A pear-shaped thick muscular organ
10. Non-gravid uterus lies in the lesser pelvis with its body resting on bladder
11. cervix between the bladder and rectum
12. Anteverted (relative to the vagina) and anteflexed (relative to the cervix)
    a. Disposition can cause uterine prolapse
6. Body of the uterus
    a. Superior 2/3 of the uterus
    b. Fundus
    c. Within broad ligament (mesometrium)
7. Isthmus of the uterus: constricted segment demarcating the body from the cervix
8. Cervix of the uterus
    a. The cylindrical inferior 1/3 of the uterus
    i. Supravaginal part
    ii. Vaginal part
    iii. External os of the uterus
    iv. Vaginal fornix: a narrow recess surrounding the vaginal part of the cervix, the posterior part is the deepest
9. Cervical canal:
    a. The lumen of the cervix of the uterus
    b. From the internal os to the external os
    c. Pap smear to obtain tissue and detect premalignant conditions

R. Uterine support

1. All work to prevent prolapse of internal organs
    a. Pelvic diaphragm, tonic contraction provides active support
    b. Uterine ligaments (passive support):
        i. Broad ligament of the uterus
        ii. Suspensory ligament of the ovary
        iii. Round ligament of the uterus
        iv. Cardinal (transverse cervical) ligaments
        v. Uterosacral ligaments

Vagina

1. A distensible musculo-membranous tube
2. Extending from the middle cervix to the vaginal orifice
3. Serving as an excretory channel for menstrual fluid
4. Birth canal
5. Sex organ
6. Relationships
    a. Anteriorly: fundus of the urinary bladder and urethra
    b. Posteriorly: anal canal, rectum, and recto-uterine pouch
    c. Laterally: levator ani, and ureters

S. Arterial supply of the Ovaries, Uterus and Vagina

9. Ovary and uterine tube: Ovarian and tubal branches of both ovarian artery and uterine artery
2. Anastomoses
3. Uterus:
    a. Uterine arteries

4. Vagina:
    a. Superior part: descending branch of uterine artery
5. Middle part: vaginal artery
6. Inferior part: branches of internal pudendal artery

E. Innervation of the Ovaries, Uterus and Vagina

1. Sympathetic innervation from T12- L2/3
2. Parasympathetic innervation from Pelvic splanchnics
3. Sensory fibers follow pelvic pain line conventions
4. Vagina sensory
    a. Superior 3/4: autonomic sensory to S2-4
    b. Inferior is somatic via deep perineal branch of pudendal
    c. Inferior ¼: deep perineal nerve
5. This is clinically significant since it offers patients different types of anesthesia for childbirth.

F. Lymphatic drainage

1. Round ligament to superficial nodes
2. Ovary and distal tubes to aortic/lumbar
3. Vagina to sacral and internal iliac
    a. Portion in vulva to superficial
4. Cervix to superficial inguinal and iliac
5. Uterus is varied

G. Obstetric anesthesia

1. Epidural into L3 epidural fat
    a. Preplanned
    b. Catheter allows for readministration
    c. Numbs birth canal and perineum
    d. Does not travel to brain
2. Spinal into subdural space
    a. Everything from the waist down
    b. Short lasting
    c. Usually stays in subdural spaces
    d. Can travel to cerebral spaces and cause headaches
3. Pudendal is a peripheral nerve block
    a. Into perineum
    b. Numbs perineum and distal vagina
    c. Allows patient to feel contractions
    d. Can be readministered
    e. Ischial spine is landmark

Answers to Study Questions

These answers are not meant to be all-encompassing, rather they are examples of the types of thought processes needed for application of the material.

1. What is the most common site of ectopic gestation? And discuss.
    A. Abdominal cavity
    B. Ampulla of uterine tube
    C. Cervix of uterus
    D. Infundibulum of uterine tube
    E. Isthmus of uterine tube

Answer: Although ectopic pregnancy can occur at various sites such as peritoneal cavity, ovary and any part of the uterine tube, the most common site is the ampulla of the uterine tube. Over 95% of the ectopic pregnancies occur in the uterine tube and mostly in the ampulla. Ectopic pregnancy results when the passage of a fertilized ovum along the uterine tube is impeded, and the morula is implanted outside of the endometrium. Inflammatory diseases of the uterus and/or vagina can lead to mucosal adhesion or/and abnormal tubal motility, eventually occlusion of the uterine tube.

2. Review the course of the vas deferens and discuss the normal site for the performance of a vasectomy.

Answer: The ductus deferens (vas deferens) continues with the duct of the epididymis at the inferior pole of the testis. It ascends posterior to the testis and medial to the epididymis in the scrotum as the primary component of the spermatic cord. The vas deferens has a thick muscular wall and a minute lumen, thus appears as a firm cord during palpation. The vas deferens passes through the anterior abdominal wall via the inguinal canal, crosses the external iliac vessels and enters the pelvis through the deep inguinal ring lateral to the inferior epigastric vessels. In the pelvis, it travels along the lateral wall of the pelvis, external to the parietal peritoneum.

Near the posterolateral angle of the urinary bladder, the vas deferens crosses superior to the ureter to reach the fundus of the bladder. The relationship of the vas between the ureter is similar, although of lesser clinical importance, to that of the uterine artery to the ureter. Posterior to the bladder, the vas deferens at first lies superior to the seminal gland and then descends medial to the ureter and the

gland. Here, the vas deferens enlarges to form the ampulla of the vas deferens and then merge with the duct of seminal vesicle to form the ejaculatory duct.

A vasectomy is a common procedure of sterilization. During this procedure, part of the vas deferens is ligated and/or excised through an incision in the superior part of the scrotum. Thus, the subsequent ejaculated semen contains no sperms.

3. A 32-year-old patient presents with gonorrhea and a tubo-ovarian abscess. Discuss the implications of passage of the bacteria through the vagina and uterus.

Answer: Sexually transmitted infections of the reproductive tract are common and can be caused by many organisms i.e., *Neisseria gonorrhoeae*. The bacteria can ascend through the vagina, cervix and enter the uterine cavity. In the uterine cavity, they can attach to the mucosal cells and cause acute endometritis. The bacteria proceed to the uterine tube and elicit acute inflammation, acute salpingitis. Infection may spread to the ovary, pelvic and abdominal cavity through the abdominal orifice of the uterine tube, causing tubo-ovarian abscess, subdiaphragmatic and pelvic abscesses.

4. Compare and contrast the anesthesia options available during childbirth and describe the anatomic basis for each type.

Answer:

1. *General anesthesia:* the patient is unconscious and unaware of the labor and delivery. They won't be able to participate actively in childbirth and miss the earliest moments of the child's life.
2. *Spinal anesthesia:* anesthetist agent is introduced into the spinal subarachnoid space at the L3-L4 vertebral level. The perineum, pelvic floor, and birth canal, and the entire lower limb are anesthetized. The patient is conscious, but the sensation of the uterine contractions is also eliminated. The spinal subarachnoid space communicates with the cerebral subarachnoid space. When the patient lies flat after delivery, the remnant of the anesthesia agent can be circulated into the cranial cavity and causes severe headache.
3. *Caudal epidural block:* a popular choice. The anesthetic agent is administered in the sacral canal using an in-dwelling catheter. In an extended labor, more anesthetic agent can be administered for a more prolonged anesthesia. In the sacral canal, the S2-S4 spinal nerve roots are blocked. That includes the visceral pain fibers from the uterine cervix and superior vagina, somatic pain fibers in the pudendal nerve that innervating the inferior part of the vagina and the

perineum. Thus, the entire birth canal, the pelvic floor, and perineum are anesthetized. Most of the lower limbs are spared. The pain fibers from the uterine body are not affected and the patient is aware of the contraction of their uterus.

4. *Pudendal nerve block*: only the perineum and inferior part of the vagina are blocked.